ADAPTIVE AND EXPLAINABLE HUMAN-ROBOT COLLABORATION FOR INTELLIGENT SORTING SYSTEMS

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COLLABORATIVE TEXTURE RECOGNITION

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DECLARATION

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ABSTRACT

The combination of artificial intelligence with robotics has been revolutionary in increasing human–robot collaboration (HRC). This research is aimed at improving the capability of HRC through developing a collaborative texture recognition module under an adaptive HRC architecture. This will be achieved using the educational robot known as "Woody". The aim is to enable the accurate classification of different material textures such as plastic, glass, rubber, and metal with an emphasis on providing an inclusive learning experience for disabled learners.

For the purpose of this research, several machine learning approaches have been investigated with the aim of finding the best texture classification method. Vision-based deep learning models as well as sensor-based approaches were employed. These include Sequential CNN, EfficientNet, and MobileNet that achieved an accuracy of 0.44, 0.88, and 0.89 respectively. The final model used DINOv2, which is a vision transformer that attained an accuracy of 0.98. On the other hand, capacitive and ultrasonic sensor-based inputs were examined to increase environmental awareness. The former failed to detect certain non-metallic materials thus making ultrasonic sensors the preferred method to achieve inclusive learning.

It is meant to be used as an assistive learning aid and is under consideration for implementation in partnership with SERD Kelaniya, demonstrating the practicality and societal value of the solution in educational settings. The experimental findings confirm the efficiency of the suggested hybrid method, showing that a combination of modern vision models with sensory inputs greatly enhances their performance in practice.

The framework not only supports intelligent automation but also promotes inclusivity in education by bringing modern robotics technology into human learning applications.

Keywords: Human–Robot Collaboration, Texture Recognition, DINOv2, Assistive Learning, Material Classification

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Table of Contents

DECLARATION i
ABSTRACT ii
ACKNOWLEDGEMENTS iii
LIST OF FIGURES vii
LIST OF TABLES viii
LIST OF ABBREVIATIONS ix
1. INTRODUCTION 1
1.1. Background & Literature survey 1
1.1.1. Background 1
1.1.2. Literature survey 3
1.2. Research gap 5
1.3. Research Problem 7
1.4. Objectives 9
1.4.1. General Objective 9
1.4.2. Sub-Objectives 10
1.4.2. Outcome Alignment 11
2. METHODOLOGY 12
2.1. Methodology 12
2.1.1. Overview of the System 12
2.1.2. System Architecture 13
2.1.3. Dataset and Data Acquisition 16
2.1.5. Sensor-Based Perception Module 18
2.2. Commercialization Aspect 21
2.2.1. Target Market and Application Domain 22
2.2.2. Value Proposition 23
2.2.3. Scalability and Future Commercial Potential 23
2.2.4. Social and Educational Impact 24
2.2.5. Summary 24

2.3. Testing and Implementation 25
2.3.1. System Implementation Environment 25
2.3.2. Vision Model Training and Evaluation 25
2.3.3. Sensor System Evaluation 26
2.3.4 Robotic System Testing 26
2.3.5. Real-Time System Integration Testing 27
2.10.6. Performance Evaluation Summary 27
3. RESULTS AND DISCUSSION 28
3.1. Results 28
3.1.1. Vision Model Performance 28
3.1.2. Sensor System Results 29
3.1.3. Robotic System Performance 30
3.1.4. Integrated System Performance 31
3.2. Research Findings 31
3.2.1. Effectiveness of Transformer-Based Models 31
3.2.2. Limitations of Classical CNN Architectures 32
3.2.3. Importance of Multimodal Sensor Fusion 32
3.2.4. Limitations of Capacitive Sensing 33
3.2.5. Educational Impact of the System 33
3.3. Discussion 34
4. CONCLUSION AND FUTURE WORK 37
4.1. Conclusion 37
4.2. Contributions of the Research 38
4.3. Limitations of the Study 38
4.4. Future Work 39
4.4.1. Expansion of Material Categories 39
4.4.2. Integration of Advanced Sensor Technologies 39
4.4.3. Improvement of Robotic Intelligence 40
4.4.4. Real-Time Edge Deployment 40
4.4.5. Enhanced Assistive Learning Features 40

4.4.6. Multi-Robot Collaborative Systems 40
4.5. Final Remarks 41
REFERENCES 42

LIST OF FIGURES

Figure 3.2: Texture recognition diagram 1.....**Error! Bookmark not defined.**

LIST OF TABLES

Table 3.1: Performance of models 1 28

LIST OF ABBREVIATIONS

Abbreviation	Description
HRC	Human Robot Collaboration
AI	Artificial Intelligence
CNN	Convolutional Neural Network
ResNet	Residual Network
RL	Reinforcement Learning
NLP	Natural Language Processing
XAI	Explainable Artificial Intelligence
ML	Machine Learning
HRI	Human Robot Interaction
BERT	Bidirectional Encoder Representations from Transformers
PCA	Principal Component Analysis
YOLO	You Only Look Once

1. INTRODUCTION

1.1. Background & Literature survey

1.1.1. Background

Rapid advancement in AI, Robotics, and Machine Learning has significantly impacted modern automation systems, as machines have become more advanced and able to perceive, think, and manipulate with increased autonomy and efficiency. One notable paradigm that has been recently introduced in this context is HRC, which involves cooperation between human beings and robots working in the same environment in order to complete tasks together. As opposed to conventional industrial automation systems that work independently, the emphasis in HRC systems is on adaptability, safety, and interaction with dynamic environments [1], [2].

Robotic sorting systems fall under applications of HRC that need real-time object perception and manipulation. Object recognition and detection methods using deep learning such as YOLO have gained popularity as they can detect objects at a very fast pace in real-time [3]. Moreover, RGB-D perception systems are used to assist the robotic system to perceive its environment in terms of visual and spatial understanding [4].

Despite, the problem of recognizing materials or textures is still one of the difficult problems faced in robotic vision perception. It requires high-level analysis to distinguish between plastic, glass, rubber, and metals, among others, based on their properties, reflectivity, and textures. Current CNN models typically fail in different lighting conditions, object occlusions, and noises present in an environment [5] [17].

However, recent developments in deep learning algorithms are offering solutions using transformers and self-supervised learning. The use of vision transformers without a massive amount of labeled data has shown a great ability to learn the semantic representation of images [6]. Specifically, DINOv2 has been shown to learn highly transferable and robust visual representation, thus making it very useful for fine-grained vision tasks like material and texture recognition [7].

The use of sensor-based perception systems adds more intelligence to robots with the help of additional non-visual sensing methods. Multisensory approaches based on integrating visual sensory inputs with other types of sensors like capacitive sensors,

ultrasonic sensors, and proximity sensors have helped robots gain better knowledge of their surroundings [8]. However, single sensory input alone does not provide adequate data in complicated real-life settings; hence, there is a trend towards sensor fusion [9].

Robotic systems require real-time operations. This is why edge computing has been adopted to allow direct processing of sensor and visual inputs in robot systems [9].

Human-robot collaboration has also been extensively studied in industrial environments and shared workspaces. Adaptability, safety measures, and human intentions have been found to be vital components for successful collaboration [10], [11]. Further, adaptive control systems in robots can enable them to modify their behavior according to human actions [12].

Besides being applied to industrial environments, robots have been increasingly used in other fields, such as assistive robotics and educational robotics, especially for learners with disabilities. Assistive robotic devices that incorporate AI technology have shown a lot of promise for personalized learning [13]. Socially assistive robots offer another dimension of assistance by engaging learners in an interactive learning environment [14], [15].

Intelligent tutors combined with robotics can add even more value to learning by offering multimodal feedback, adaptive instruction, and interaction, especially within the scope of special education [16]. Such approaches demonstrate how crucial it is to combine perceptual capabilities, interaction, and adaptation in robotic systems that are used in an educational environment.

Nevertheless, the lack of research in this field demonstrates that most of the current solutions are concentrated either on efficient perception models or learning assistance algorithms separately. Moreover, there is little research that combines such advanced techniques as self-supervised visual learning models, represented by DINOv2, and learning assistance robotic systems using multimodal sensors.

1.1.2. Literature survey

There have been several studies related to Human-Robot Collaboration (HRC). According to Ajoudani et al., there is an evolutionary trend of HRC where adaptability, safety, and efficiency are essential aspects [1]. Moreover, Lasota et al. [2] offer an overview of the methods that allow safe interactions between humans and robots, especially when it comes to perception and the ability of avoiding collisions. Similarly, the problem of adaptive robot collaboration, which is the ability of a robot to adjust its behavior according to actions performed by a human, is addressed by Vasic and Billard [12].

Robotic perception mostly relies on the use of deep learning algorithms. Object detection based on YOLO architecture is very popular due to high efficiency and accuracy [3]. Moreover, RGB-D perception algorithms significantly contribute to robotics since they use depth and RGB streams simultaneously [4].

The most widespread algorithms used for recognition and sorting of materials include CNNs. However, the problem of generalization is still relevant since even the state-of-the-art methods are not fully capable of dealing with real-world cases. Therefore, transformer-based models are increasingly employed nowadays.

There have been remarkable advancements made with self-supervised learning in visual representation learning. Caron et al. [6] have proposed vision transformers that can learn useful visual representations without any labels. Furthermore, DINOv2 has outperformed all existing methods with regard to fine-grained visual comprehension and is very well-suited for texture and material recognition [7].

Sensor fusion has emerged as an important way to increase the reliability of robotic perception. Xu et al. [8] state that by integrating information from several sensors, it is possible to make robots work reliably even in complex environments. Additionally, Melgarejo-Sánchez et al. [9] prove that edge computing facilitates efficient real-time processing of multisensor data in robotic platforms.

On the other hand, there have been some developments in the area of adaptive robotics and modeling human behavior. Chen et al. [17] and Rossi et al. [18] state that it is necessary to understand the intentions of humans in order to collaborate effectively.

The advantages of using AI-enabled systems in assistive and educational robotics are numerous for promoting inclusive education. According to the study of Ijriss and Cabanes [13], assistive technologies help increase accessibility and personalize education for students with disabilities. The study conducted by Nadeem et al. [14] presents a literature review of socially assistive robotics emphasizing advancements in engagement and communication. The role of social robots in increasing motivation and learning results is confirmed in the study by Belpaeme et al. [15].

It has been found that integration of robotics in intelligent tutoring systems leads to greater effectiveness through the provision of adaptive feedback and multimodal interaction [16]. These systems prove to be especially helpful in special educational settings where cognitive engagement can be achieved through structured interactions.

As far as applied research into robotics is concerned, there are several recent examples to consider. For instance, Chinchmalatpure et al. [19] discuss IoT-based YOLO-driven object sorting systems, whereas the study by Wang and Peng [20] focuses on robotic manipulation systems to solve object handling problems.

However, in spite of these developments, existing literature primarily deals with one of the two, that is, advanced perception algorithms or interactive learning robots, separately. Little literature is available on the use of contemporary self-supervised vision algorithms like DINOv2 alongside multimodal sensor fusion techniques and assistive robot learning algorithms. This highlights an area of considerable research interest regarding the development of intelligent adaptive and inclusive robotics systems.

1.2. Research gap

Despite many developments in human-robot cooperation, robotic perception, and assistive robotics, there are still several important limitations in existing studies. The majority of HRC architectures pay little attention to detailed visual perception necessary for recognizing objects by their material properties [1], [2]. Even though collaborative control enhances interactions in human-robot collaborations, it does not solve problems related to intelligent perception in unstructured and dynamic settings [10], [12].

Regarding material recognition in robotic sorting systems, deep learning techniques such as YOLO perform very well in terms of object localization [3]. Yet, these methods are mainly focused on identifying objects without paying much attention to their textures or material properties that become crucial when it comes to recognizing similar materials such as plastics, rubbers, and glasses [5]. Besides, while RGB-D perception can enhance spatial perception, it is still insufficient for recognizing materials depending on different light conditions and surfaces [4].

The development of self-supervised learning and vision transformers has led to a remarkable improvement in visual representation learning techniques. The models proposed by Caron et al., and DINOv2 show great potential in visual representation learning with excellent generalization ability [6], [7]. Yet, the implementation of these models within actual robotics applications along with physical sensing and human interactions has not been thoroughly researched, especially in educational or assistive applications.

In the domain of sensory fusion, researchers have focused on incorporating vision and physical sensing to boost robot perception [8]. In spite of this, most available research only focuses on simple sensor fusion without integrating vision and intelligent learning techniques. Moreover, edge computing has provided an efficient solution for enhancing real-time computation in robots; however, it seldom incorporates advanced vision transformers into decision-making tasks [9].

Concurrently, technologies like assistive robotics and AI-driven education tools are receiving growing interest in helping differently abled students learn better. It is evident from research that assistive socially interactive robots and intelligent tutors enhance engagement, communication, and learning outcomes for special education environments [13], [14], [15], [16]. Yet, these technologies are geared toward aiding interactions and behavior changes, and not towards material interactions.

Table 1.1: Comparative Analysis of Existing Approaches

Study Area	Existing Approaches	Limitations Identified	References
Human–Robot Collaboration	Industrial task coordination, safety-focused frameworks	Limited intelligent perception and educational adaptation	[1], [2], [10], [12]
Object Detection in Robotics	YOLO, CNN-based real-time detection	Weak fine-grained material/texture classification	[3], [5]
Vision-Based Perception	RGB-D and CNN-based perception systems	Sensitive to lighting, occlusion, and texture variations	[4], [5]
Self-Supervised Learning	Vision Transformers, DINOv2	Limited real-time robotic and sensor integration	[6], [7]
Sensor Fusion in Robotics	Vision + physical sensor combinations	Mostly rule-based, limited deep fusion learning	[8], [9]
Assistive Educational Robotics	Social robots, tutoring systems	Focus on interaction, not physical object recognition	[13], [14], [15], [16]
Robotic Educational Systems	Basic manipulation and demonstration robots	Lack of perception + adaptive learning integration	[19], [20]

Table 1.1 provides a concise overview of the major areas of research, the approaches adopted by various researchers, and their limitations. As can be seen from the above table, although there have been tremendous advancements in specific fields such as human-robot cooperation, perception through vision, and assistive robots, none of them have been able to integrate the various aspects. The comparison makes it evident that there is a need for a system that encompasses all the three approaches mentioned.

There are some essential gaps identified by the research analysis:

- No coordination between the self-supervised vision model and the perception systems of the robot.
- Lack of multimodal combination between vision and sensors along with human-robot collaboration mechanisms.
- Minimal attention on precise material classification for robotic sorting applications.
- No existence of real-time educational robots that are able to perform object recognition and learning activities.
- Lack of systems that have an integrated vision model, interaction system, and learning mechanism.

Hence, it can be concluded that there should be a requirement of such an intelligent robotic system that has an efficient self-supervised vision model like DINOv2, along with multimodal sensory fusion and human-robot collaboration.

1.3. Research Problem

Although considerable advancements have been achieved in human-robot collaboration (HRC), robotic perception, and educational assistance using technology, one major drawback is still the inability of current technologies to be able to develop robotics that will have high accuracy material recognition capability, real-time multimodal perception, and effective interactions in educational environments.

Current HRC architectures are mainly geared towards industrial purposes where safety, efficiency, and cooperation of human operators and robots are key considerations [1], [2], [10]. Although they make possible efficient collaboration, they fail to take into account the fine perception of physical objects necessary for practical object-based learning and manipulation tasks.

Robotic perception, on the other hand, has made remarkable achievements through deep learning methods such as the popular YOLO framework. These models have achieved efficient real-time detection of objects [3]. They only fail in distinguishing between visually similar materials like plastic, rubber, glass, and metal depending on the environment they find themselves [5]. RGB-D based perception makes robots have better spatial awareness but fails in texture-based classification [4].

Though there is evidence of high capability of learning transferable visual representation by recently proposed self-supervised vision transformers like DINOv2 [6], [7], the implementation of such models in robotic systems that leverage sensor fusion and human-robot collaboration architectures has received little attention. The current applications mainly focus on off-the-shelf solutions and vision-only problems, making them less useful for robot-human interactive applications.

Alternatively, sensors and sensor-fusion strategies have been applied for perceptual improvements through a combination of vision and physical sensing [8], [9]. But these robotic systems typically adopt rule-based mechanisms or loose fusion methods where the capabilities of deep learning methods for informed decision-making were not utilized for accurate classification.

In the domain of assistive and educational robotics, socially assistive robots and intelligent tutors have been found to be effective at increasing engagement and learning outcomes of differently-abled students [13], [14], [15], [16]. But the current systems lack focus on perception and classification of objects, thus being ineffective in practical hands-on training scenarios.

Hence, the primary research problem investigated in this study is:

How can a human-robot cooperative design that combines state-of-the-art self-supervised perception systems for vision, as well as sensor fusion for multimodal information processing, be constructed so that material classification can be accomplished in real-time, and at the same time, an interactive and learning experience can be enabled for differently abled children?

The key focus of the research problem involves creating a solution that would integrate all four aspects:

- High-level perception systems (DINOv2-based perception)
- Low-level sensor fusion (Ultrasonic/proximity sensing)
- Human-robot cooperation methods
- Educational assistance

The main goal of the research problem investigation is a system that could successfully combine these two features – material recognition as well as education

1.4. Objectives

The primary focus of this research is on developing an adaptive human-robot collaboration system (Woody educational robot), using state-of-the-art deep learning-enabled vision models along with multimodal sensory fusion technologies for accurate real-time classification of material textures, and providing interactive learning for children with special needs.

This research seeks to create an adaptive human-robot collaboration system that addresses the problem area related to robotics, intelligence, and education together by bridging the gaps among all three of these fields.

1.4.1. General Objective

- To design a framework for effective cooperation between humans and robots that incorporates self-supervised visual learning along with sensor perception to accurately classify material textures in real time within an assistive educational setting.

The goal is to create a unified human-robot collaboration framework which uses self-supervised vision learning as well as sensor perception techniques to accurately and reliably classify textures of everyday objects/materials, specifically plastic, glass, rubber, and metal, in real-time.

The goal emphasizes the ability of the system to recognize materials based on their visual characteristics, such as appearance, reflective properties, transparency, etc., which cannot be accomplished through ordinary methods of object detection. In addition to processing the information obtained from real-world environments through RGB image sensors, the system would be able to classify materials regardless of environmental factors like lighting and orientation.

In addition to material recognition capabilities, another key component of this objective is creating the system within a human-robot collaboration framework to

enable safe interactions between the robot and users. This allows the robotic system not only to identify materials accurately, but also use the identification process as an element of user-friendly experience in which material properties can be taught by showing users visually observable differences and properties.

1.4.2. Sub-Objectives

1. For designing and developing the material recognition system using vision

For the development of the material recognition algorithm based on deep learning using various types of materials such as plastics, glass, rubber, and metals. Convolution and Transformer models were used in order to classify material textures using RGB images collected from real-life scenarios.

2. Comparing different deep learning models for optimal classification performance

For testing and comparing various deep learning models, including sequential CNNs, EfficientNets, MobileNets, and DINOv2s for material texture classification. This comparison study was conducted to determine which model can be considered efficient based on performance and accuracy. It was discovered that the DINOv2 architecture outperformed other models with 0.98 accuracy in classification tasks.

3. To implement multimodal sensor fusion

For improving the efficiency and effectiveness of the multimodal approach by implementing both visual and physical sensors for object perception. At first, a capacitive proximity sensor was implemented in order to detect non-metallic objects. However, the experiment did not yield satisfactory results. Therefore, an ultrasonic sensor was implemented to increase perception reliability.

4. To develop a real-time robotic control system for collaborative operation

Design and development of robotic arm control system employing MG996R servo actuator-based sorting of objects according to the output of a pre-trained model. The control system facilitates real-time collaboration between the operator's inputs and the decisions made by the robot to safely sort objects in a shared workspace.

5. To develop the system as an assistive educational system for differently-abled students

To turn the developed robot into an assistive system in teaching students with disabilities about the process of sorting materials. The developed robotic arm will provide an interactive learning experience for the students as they observe and learn the process of sorting material in real time. This system can be used with the help of SERD Kelaniya for use in special education institutes.

6. To assess the accuracy and usefulness of the system in a real-world scenario

To test the accuracy of the complete system including classification, sensory inputs, and robot control operations in real-time. Accuracy testing shows the proposed system performs well with a classification accuracy of approximately 98% with the use of DINOv2-based vision model.

1.4.2. Outcome Alignment

The successful accomplishment of the research objectives led to the following outcomes:

• The creation of a high-precision material classifier model using the DINOv2 algorithm, with a real-world test accuracy of about 0.98

• A complete multimodal robotic sorter system that combines visual perception and sensors for accurate operations

• The creation of an educational robot system that helps in creating an inclusive environment where children with disabilities can learn

• Better knowledge regarding the integration of artificial intelligence-powered perception algorithms with robots in practical settings

• An effective framework for human-robot interactions in assistive education environments

2. METHODOLOGY

2.1. Methodology

2.1.1. Overview of the System

This suggested system will be an adaptive platform for collaboration between humans and robots named "Woody", which is created to provide material texture classification and robotic sorting with intelligence in real time. The purpose of developing such a system is associated with bridging the gap between artificial intelligence, robotics, and educational process by making an effort towards involving differently-abled students in this process through interactivity and experience of collaboration.

While classic robotic sorting platforms are focused mainly on the automation process within the framework of industry needs, the Woody system is based on an approach which implies involvement of human participants in the process. Namely, the robotic arm does not simply complete the task of sorting, but acts as an interactive assistant to the learner who can understand the process of perception of object's appearance by the system, analysis and decision-making process carried out by the underlying artificial intelligence system as well as translation of these decisions into actions executed by the robotic arm.

The system revolves around four components that are interlinked in terms of functionality. First, there is the deep learning mechanism, followed by the sensor-based perception system, a robot actuator component, and finally, the human-robot interface. The entire system runs sequentially. For instance, the first stage starts when an object is detected via the camera. In particular, a series of preprocessing steps are performed to facilitate object classification within the deep learning framework. The sensor-based mechanism ensures that the system receives additional data to detect objects successfully and operate effectively. Lastly, once object classification is completed,

the robotic mechanism executes the sorting process through programmed servo actions.

It should be noted that the central aspect of this research is the ability to categorize objects into four primary types of material: plastic, glass, rubber, and metal. These materials have been chosen because they are ubiquitous in daily life and play an important role in industrial applications and teaching purposes. Importantly, these materials are differentiated based on specific features such as surface texture, reflectance, and transparency.

For the system to function effectively, a blend of deep learning and sensor fusion methods has been integrated. The system also prioritizes real-time operations, meaning that the system will be able to perform its tasks immediately after the user's intervention and input objects.

In conclusion, the Woody system is an innovative integration of intelligent perception, robot manipulation, and human interaction, whereby a mere automation task has been converted to education by leveraging the power of technology.

2.1.2. System Architecture

The system architecture is designed based on modular, layered, and hierarchical principles, wherein each module is dedicated to a particular role but together creates a coherent perception-decision-action sequence. This architectural strategy guarantees flexibility, scalability, and easy integration, making it possible to update or improve any particular module without compromising the efficiency of the entire system.

At the input stage, the system retrieves data in real-time through two main channels: an RGB camera and an ultrasonic sensor. The RGB camera uses the IP webcam application running on a smartphone connected to the robot. This method allows us to obtain reliable visual information for further machine learning at a relatively low cost. An IP webcam generates video footage of the objects put into the robot’s operating field. This footage is rich with crucial visual data like texture, color distribution,

reflectance, edges, and pattern information, which makes it easier to distinguish between materials.

In addition, the ultrasonic sensor works as a measuring distance and object detecting module. It constantly sends ultrasonic waves and computes the distance of objects that can be found close to the sensor through their reflection back. This helps avoid processing unnecessary objects, as well as those that are not really there. Thus, the implementation of this device increases awareness in the environment and avoids processing data when no objects are present.

Prior to classifying the object, the acquired image data goes through a preprocessing pipeline meant to optimize the input. Among other things, images are resized to fit into the resolution used in the deep learning model. Also, normalization procedures are implemented to prevent possible distortion of data. At the same time, when the data is being trained, additional operations are performed to increase the volume of images artificially. Such operations include rotating, horizontal flipping, zooming, and brightness adjustment.

The preprocessed images then proceed to the feature extraction and classification section, which is the most intelligent section of the system. During the development stage of this project, multiple deep learning algorithms have been applied and compared. These include the sequential Convolutional Neural Network (CNN) model, EfficientNet model, and MobileNet model. All three models were selected because of their efficiency in processing images and various degrees of complexity. However, after conducting experiments, it became clear that those models faced difficulties when it came to capturing differences in fine-grained texture features, which was essential for proper material recognition.

In order to address that problem, the final model chosen is DINOv2, which is a state-of-the-art self-supervised vision transformer algorithm. The primary advantage of this algorithm over the aforementioned CNN-based models is that DINOv2 allows for the extraction of global contextual information from images. Consequently, the model can differentiate between certain types of textures and materials, even when they cannot

be easily distinguished by human eyes. For example, DINOv2 would be able to differentiate between reflective and matte surfaces as well as transparent ones.

After processing through the classification algorithm, the output is fed into the decision fusion stage. At this point, the system combines visual outputs with sensor inputs to create an optimal decision-making process. The ultrasonic sensor is highly significant at this stage as it verifies the existence of objects prior to any robotic operations being carried out. Such an approach increases the accuracy of actions being taken as it eliminates unnecessary movements and errors from noise, occlusion, and accidental triggering factors that might interfere with decision-making.

The movements of the robotic arm take place in a systematic procedure that ensures classification and interactive learning happen. Firstly, the robotic arm lifts the object from the input region and puts it at a central observing point. At this time, the vision model conducts material classification based on the images taken.

After the process of classification has been achieved, the robotic arm does not sort the object but moves it to the front face such that the student can see it. This is done purposely to encourage interaction between the student and the object for the purpose of learning. The student should observe the object and determine the material it is made of (plastic, glass, rubber, or metallic). This should be done by selecting from the available choices.

In cases where the student chooses the correct answer, a feedback of Correct is displayed to show the correctness. Otherwise, feedback could be given in case of an incorrect choice.

Other than the above components, the architecture further consists of the human-robot interaction component, which will enable the users, especially the students, to interact with the system. Through this component, the system will not only be automated in carrying out sorting processes based on artificial intelligence algorithms but will become an interactive system for the users to learn how such artificial intelligence decisions lead to actions taken by robots.

In summary, through the above components, the entire architecture provides a total solution in terms of the pipeline from collecting raw data from the real world, processing it using intelligence algorithms and finally interacting with real robots.

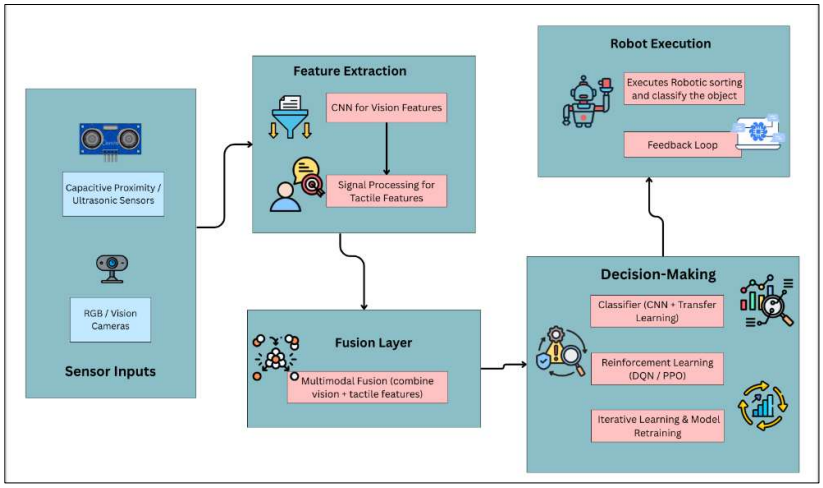

Figure 2.1: System Architecture Diagram 1

2.1.3. Dataset and Data Acquisition

In the current research, the image dataset comprises five main categories of materials: plastic, glass, rubber, metal, and "other" material. Such materials are selected based on their relevancy and appropriateness for practical sorting tasks and for purposes of teaching classification algorithms. The first four categories refer to known material types that can be encountered regularly. In turn, the "other" category implies the presence of different objects and shapes, which cannot be unambiguously classified based on their material type. It allows increasing the efficiency of material classification by accounting for unusual inputs.

The data collection process involved obtaining images in a realistic but controlled environment. In particular, images were obtained using an RGB camera, which was created using an IP Webcam application. In such a way, the process of capturing required images could be performed without using special equipment. Moreover, the use of IP Webcam allowed changing the position of the camera and providing various images taken in different situations and under varied conditions. Particularly, the data collection process took place under different levels of light, various backgrounds,

positions of objects relative to the camera, and distances from objects to the camera lens.

To maintain diversity and generalize the system, several images were taken at varying angles of each item. These included top shots, side shots, and slightly angled views. These differences ensure that the neural network does not get accustomed to recognizing objects from a particular angle but learns about their unique properties, such as surface texture, reflection, translucency, and pattern structure. For instance, reflective properties of metals and translucency of glasses require learning of more advanced characteristics than mere color discrimination.

The dataset also contains differences in size, shape, and surface quality of items, which can be either shiny, matte, smooth, or rough. This is especially crucial for texture-based recognition, as plastic and rubber can look visually alike in some cases. In this regard, by introducing variations into the training process, the neural network will learn how to rely on subtle texture differences between materials.

In order to prepare the data for the training process, every picture in the set is preprocessed in a series of steps. First of all, each image is uniformly resized to fit into the required resolution that corresponds to input requirements of the neural networks. Pixel intensities are normalized to the same range which makes the learning process more stable and helps converge faster. In addition to that, noise reduction may be used in certain cases as an extra step.

To augment the number of samples, various techniques of data augmentation are utilized during the training process. They may consist in flipping the image horizontally, rotating it slightly, changing its brightness level and zooming. Such manipulation not only enlarges the dataset but also helps to avoid overfitting by adding diversity to it.

As we can see from above, our dataset is created with care so that it is diverse enough and realistic enough to allow the neural network to extract important information from the image. The quality of data is crucial as it influences the results greatly.

2.1.4. Vision-Based Material Classification Module

The visual classification module is arguably the most crucial part of the entire system as it handles the process of analyzing raw images and predicting materials based on their appearance. At first, classic convolutional neural network structures were utilized as base models. The Sequential CNN structure was employed as the most straightforward method of extracting features from the image and classifying the material. Unfortunately, this model was not capable of delivering satisfactory performance, achieving an accuracy of around 0.44, which implies that it failed to distinguish between various material appearances.

To enhance the performance of the model, more sophisticated architectures such as EfficientNet and MobileNet were used. Both EfficientNet and MobileNet are examples of efficient CNN designs that combine high accuracy with low resource requirements. While the EfficientNet model delivered an accuracy score of about 0.88, the MobileNet architecture was able to achieve a result of approximately 0.89.

A transformer architecture based self-supervised learning algorithm, known as DINOv2, is proposed to alleviate such constraints. Unlike other algorithms, the training of DINOv2 does not depend on a huge amount of labeled data; rather, it creates visual representation by contrasting augmented views of the input image.

In our system, the model DINOv2 scored an accuracy of approximately 0.98, which is far better than other CNN-based models. The model was found to have great potential in recognizing fine texture details.

2.1.5. Sensor-Based Perception Module

Aside from visual classification, another layer of sensory perception has been employed in order to increase the reliability of the system and its awareness of the environment.

Firstly, the possibility of using a capacitive proximity sensor for material classification was attempted. Based on initial expectations, this particular type of sensor should have helped to recognize different materials through the use of their electric properties. Nevertheless, it was discovered that the capacitive proximity sensor could only accurately detect metallic objects; it did not work properly for other types of materials like plastic, rubber, and glass.

Thus, the capacitive sensor was substituted with an ultrasonic sensor. As opposed to the former one, ultrasonic sensors do not perform material recognition; nevertheless, they provide crucial information regarding whether there are any objects close to the robotic arm and how far away they are located from it.

The role of this sensor is rather secondary; it allows avoiding unnecessary movements of the robotic arm because it only acts when an object is actually there.

2.1.6. Robotic Control and Actuation System

As for the actuator system, the implementation of the robotic arm is done with the help of servo motors (MG996R type). These particular types of motors ensure adequate torque and stability necessary for successful operation of the system. As for functionality of the actuator system, it is not only responsible for performing actions, but also serves as the means of interactive human-robot learning.

Differently from traditional robotic sorting systems where the robot directly sorts objects into bins depending on their classifications, our solution uses multi-step interaction between user and robot to improve perception accuracy as well as enhance the process of learning.

First of all, the robotic arm moves the item from the area where objects enter the workspace to a predefined Region of Interest (ROI). This region is a specifically designed area in the center of the workspace which ensures that images captured by the camera are not impacted by background and other objects around. Isolation of objects in ROI allows ensuring high perception accuracy.

After placing the object within the ROI, the RGB camera starts capturing the object's image which is further processed by the DINOv2-based classification model.

Once the classification is done, the robotic arm enters the next stage, which is the interaction stage. Here, instead of directly sorting the item, the robotic arm relocates it to an easily accessible area for the learner. This is intended to enable assistive learning, as the learner is able to view the object and engage in the classification activity.

Here, the learner is asked to determine the material group of the object, which could either be plastic, glass, rubber, metal, or any other option. The output from the learner is compared to that obtained from the model prediction. If the classification is correct, the system offers a feedback statement saying 'Correct'. On the other hand, if the classification is not accurate, an appropriate message could be presented to the learner.

The sorting step involves mapping the classification results to a series of movements, which include picking up the object, lifting it, rotating it, and placing it at its designated location.

The entire process, from picking up the object to placing it down in the designated location, is carried out seamlessly and precisely timed to ensure effective transitioning of the phases. The utilization of servo motors makes it possible to control the angle of joint motion and the trajectory of movement.

In addition, the robot acts in real-time, meaning that its actions are responsive because there are no time delays involved in perceiving, deciding, and executing actions. The timeliness of actions is crucial in keeping interactions lively and engaging, especially when using robots in an educational setting.

In general, the robotic control and actuation system upgrades a simple sorting procedure into an interactive and engaging learning tool.

2.1.7. **Human-Robot Interaction Design**

The human-robot interaction module is one of the distinctive modules in this system compared to other conventional sorting robots. The system does not only focus on automation; it is also designed to be interactive and informative.

The users, especially the students, will have a chance to watch how objects are sorted out through this system. There is feedback provided by the system which helps the user understand how the AI model perceives various materials.

Moreover, the system is designed to be safe in an environment where both human and robots co-exist. The robot is programmed to have a safe level of motions which avoids any form of unsafe actions.

Such a system would make a perfect fit in assistive educational institutions where students who have some forms of disabilities are taught through a practical AI-enabled system.

2.2. Commercialization Aspect

Commercialization is one of the main aspects of this research which aims at developing the proposed Woody human-robot collaborative system from a laboratory prototype into a viable solution which can function in real life situations. Contrary to other robotic sorting systems which are developed mainly for purposes of automation in industries such as in warehouse sorting or in logistics management, the Woody human-robot collaborative system is designed based on a human-centered approach.

In this case, the main target audience is children with special needs for the purpose of promoting inclusive learning and education as well as accessibility. In order to promote inclusive learning, the system is intended to encourage learning in a way that promotes engagement among learners and educators in educational settings as well as in assistive learning programs.

One of the pathways which can lead to the commercialization of the system is through collaboration with SERD (Services for Empowerment and Rehabilitation of Disabled Children and Families). SERD is a specialized institution in the Western Province, Sri Lanka. It is a registered organization under the category of social services.

The mission of SERD is to ensure the equal rights of children who are differently abled as independent people or even as socialized individuals. In addition, in terms of its mission, SERD strives to integrate children who are differently abled in the regular education system, which allows these kids to participate actively in all kinds of educational activities.

Moreover, the mission of SERD includes ensuring that all actions are taken to include children with special needs in the regular social environment and that they have equal rights to life without any embarrassment. It can be noticed that this mission of SERD is in line with the mission of the Woody system because it is also designed to include children with special needs in the educational system. In addition, SERD will be helpful for testing the Woody system in real-life conditions.

2.2.1. Target Market and Application Domain

The first domain where the technology can be commercialized is the special education or inclusive learning environment. There is a need for novel tools which are required to accommodate various levels of learning capabilities of students. In such cases, the system can be utilized to help individuals who suffer from learning difficulties, autism spectrum disorders, or other cognitive and developmentally challenged conditions.

In such an environment, the Woody system will work as a tutor that provides students with an opportunity to see how the AI algorithm classifies objects in real time while interacting with them physically. Such a combination of visual and physical interaction can improve learning.

Apart from special education, another secondary domain where the technology can be applied includes STEM learning environments. Here, it will serve as a tool to demonstrate fundamental principles of artificial intelligence, machine learning,

robotics, and computer vision. Also, the technology can be applied in university laboratories or training centers where learners can get hands-on experience in working with advanced AI-robotics systems.

2.2.2. Value Proposition

This system offers considerable value by combining three aspects: artificial intelligence, robotics, and assistive education into one platform. The proposed technology uses an active and experiential approach to education rather than the traditional approach that only uses passive learning techniques where the learner simply learns about something but does not have firsthand experience in it.

Incorporation of an interactive component in the proposed system is one of the value additions because the user gets to see how the AI system works in real-time as it processes the image and determines whether the material in front of it is plastic, glass, rubber, metal, or another substance. This makes it easier for learners to understand some machine learning concepts.

The robot in our system plays a crucial role in making sure that there is an element of kinesthetic experience for the learners, as learners get to feel the movements of the robotic arm while it moves an object. This makes the proposed technology more engaging for users.

Technically speaking, our model has shown excellent results (around 98%) in terms of classification accuracy. Besides, our system utilizes multimodal sensing, which uses vision along with ultrasonic input.

2.2.3. Scalability and Future Commercial Potential

The modular design of the system makes it highly scalable for future development. Each core component—vision processing, sensor fusion, and robotic control can be independently upgraded or extended without redesigning the entire system.

Future commercialization pathways include integration with cloud-based AI platforms, enabling centralized model updates and remote learning capabilities across multiple institutions. The system can also be extended into multi-robot collaborative

learning environments, where multiple robotic units demonstrate more complex educational scenarios.

Another potential direction is the development of a low-cost educational robotics kit, making the technology more accessible to schools with limited resources. This would significantly increase adoption in underprivileged and rural educational settings.

The use of advanced transformer-based models such as DINOv2 ensures that the system remains future-proof and adaptable to emerging improvements in AI-based perception systems.

2.2.4. Social and Educational Impact

Beyond commercial value, the system has significant social impact, particularly in addressing educational inequality in Sri Lanka. Many differently-abled learners face barriers in accessing interactive and technology-driven educational tools. This system helps bridge that gap by providing an inclusive, engaging, and adaptive learning environment.

By combining AI-driven perception with physical robotic interaction, the system enables learners to better understand STEM concepts while actively participating in the learning process. This promotes equal learning opportunities and enhances motivation and engagement among students with diverse learning needs.

Ultimately, the system contributes to reducing the digital divide by ensuring that advanced technologies such as artificial intelligence and robotics are not limited to industrial or research applications but are also accessible in inclusive educational ecosystems.

2.2.5. Summary

In summary, the commercialization potential of the Woody system lies in its ability to transition from a research prototype into a real-world assistive educational platform. Through collaboration with SERD, the system can be validated, refined, and deployed in inclusive learning environments where it can directly support differently-abled learners. Its combination of high-accuracy AI-based perception, multimodal sensing, and interactive robotics makes it a strong candidate for both educational deployment and future scalable commercialization in the field of assistive robotics and intelligent learning systems.

2.3. Testing and Implementation

This chapter covers the development and testing processes of the proposed Woody system for human-robot collaboration in material classification and robotic sorting operations. The assessment process mainly considers three major aspects including vision model efficiency, sensor accuracy, and robotic operation precision.

The development process was conducted under a laboratory setting to mimic the real-life application setting of the assistive educational tool. This process encompasses the development of the deep learning-based model, sensors, and robotic arm controller. Prior to their integration into the overall system framework, each module underwent testing to assess their stability and accuracy.

2.3.1. System Implementation Environment

The system was developed using a combination of hardware and software components. The vision-based model training and testing were carried out in a Python-based environment using deep learning frameworks. The robotic control system was implemented using microcontroller-based servo control logic.

The main hardware components used include:

- RGB camera for image acquisition
- Ultrasonic sensor for object presence detection
- MG996R servo motors for robotic arm movement
- Microcontroller for actuator control and sensor integration

The software pipeline includes:

- Deep learning framework for model training and inference
- Image preprocessing and augmentation modules
- Real-time classification inference system
- Hardware communication interface for robotic control

2.3.2. Vision Model Training and Evaluation

The vision-based classification system was evaluated using four different deep learning models: Sequential CNN, EfficientNet, MobileNet, and DINOv2. Each model was trained using the same dataset under consistent conditions to ensure fair comparison.

The evaluation process involved splitting the dataset into training and testing sets. The models were trained on labeled images representing four material categories: plastic, glass, rubber, and metal. After training, each model was evaluated using unseen test data to measure its generalization ability.

The results showed a clear performance difference between traditional CNN-based models and transformer-based architectures. The Sequential CNN achieved the lowest performance, indicating limitations in learning complex texture features. EfficientNet and MobileNet showed improved accuracy due to their optimized architectures, but they still struggled with fine-grained material differentiation.

The final model, DINOv2, demonstrated the highest performance with an accuracy of approximately 0.98, indicating strong feature representation and generalization capability. This confirms that self-supervised vision transformers are more effective for material texture classification in robotic environments.

2.3.3. Sensor System Evaluation

The sensor module was evaluated to determine its effectiveness in supporting the vision-based classification system.

Initially, a capacitive proximity sensor was tested for material detection. During experimentation, it was observed that the sensor could reliably detect metallic objects due to their conductive properties. However, it failed to consistently detect non-metallic materials such as plastic, rubber, and glass. This limitation made it unsuitable as a standalone solution for multi-material classification.

To overcome this limitation, an ultrasonic sensor was integrated into the system. The ultrasonic sensor does not directly classify materials but provides reliable detection of object presence and distance measurement. This helps ensure that the robotic system only performs classification and actuation when an object is correctly positioned in the workspace.

The integration of the ultrasonic sensor improved system stability by reducing false activation and ensuring synchronized operation between perception and robotic control.

2.3.4 Robotic System Testing

The robotic arm system was tested to evaluate its ability to accurately execute sorting actions based on model predictions. The MG996R servo motors were programmed to perform a sequence of controlled movements including object pickup, lifting, rotation, and placement into designated bins.

Testing was conducted by feeding real-time predictions from the DINOv2 model into the robotic control system. Each predicted material class was mapped to a specific physical location. The system successfully executed sorting actions with minimal delay between prediction and actuation.

The robotic system demonstrated stable movement control and accurate placement of objects, confirming the effectiveness of the servo-based actuation design.

2.3.5. Real-Time System Integration Testing

After individual component testing, the full system was integrated and evaluated as a complete pipeline. This included:

- Real-time image capture
- Model inference using DINOv2
- Sensor-based validation
- Robotic actuation

The integrated system was tested under continuous operation to evaluate responsiveness and stability. It was observed that the system was able to process input, classify materials, and execute robotic actions in real time with minimal delay.

The end-to-end pipeline demonstrated reliable performance in simulated educational scenarios, making it suitable for interactive learning environments.

2.10.6. Performance Evaluation Summary

The overall performance of the system was evaluated based on the following criteria:

- **Classification Accuracy:** DINOv2 achieved approximately 0.98 accuracy, outperforming all baseline models
- **System Robustness:** Sensor fusion improved reliability in object detection and system activation
- **Real-Time Performance:** The system successfully executed real-time classification and robotic actuation with minimal latency
- **Operational Stability:** Integrated system showed consistent performance during continuous testing

3. RESULTS AND DISCUSSION

3.1. Results

This chapter highlights the findings of the experiment conducted to test the efficacy of the Woody collaborative human-robot system, with emphasis being put on three main factors: texture classification through vision, sensor system testing, and robotic movement. The tests have been carried out through controlled experiments as well as realistic testing scenarios.

Four types of materials have been selected to conduct the experiments, including plastics, glasses, rubber, and metals. All of them are commonly found in either an industrial or educational setting. Deep learning systems have been employed to test how effective they can classify textures in different scenarios, such as changes in lighting, orientation, and backgrounds.

3.1.1. Vision Model Performance

The performance of different deep learning architectures is summarized in Table 3.1.

Table 3.1: Performance of models 1

Model	Accuracy
Sequential CNN	0.44
EfficientNet	0.88
MobileNet	0.89
DINOv2 (Final Model)	0.98

As seen from the results, there is a noticeable variation in terms of performance. The Sequential CNN was able to produce the lowest classification rate at 0.44, meaning that the model lacks proficiency when it comes to texture recognition as well as recognizing discriminative features for material classifications. The model performed

poorly in identifying materials that share some visual similarities like rubber and plastic.

On introducing more efficient models, such as EfficientNet and MobileNet, there was a dramatic increase in performance as compared to the Sequential CNN. The two models have improved network design and feature-extraction abilities, making them able to recognize changes in texture patterns as opposed to sequential networks.

The best performance among all the models used was achieved using the DINOv2 model, with the accuracy rate standing at approximately 0.98. This was possible due to the self-supervised architecture and transformer-based design, allowing the model to produce generalized visual representation. In comparison to CNN, DINOv2 does not only capture global information but also localized information, helping the model recognize minor differences in texture patterns.

In conclusion, it is clear that self-supervised vision transformers can provide outstanding performance in fine-grained texture recognition tasks.

3.1.2. Sensor System Results

The detecting section of the robot was tested by utilizing two distinct types of sensors, namely the capacitive proximity sensor and the ultrasonic sensor. The purpose of testing these two different types was to find out which one of the methods would prove more reliable in detecting the presence of objects and actuating the system accordingly.

The capacitive proximity sensor proved less effective in this project due to several limitations. Although it could effectively detect objects made from metallic material, it was unable to detect objects that were not made from metal materials, such as those made of plastic, rubber, and glass. Such a characteristic of this sensor made it less useful in classifying objects, considering the fact that not all objects were made of the same material.

Unlike the above sensor, the ultrasonic sensor proved very efficient in detecting the presence of objects and measuring their distance from the sensing point. Such an

attribute made it more efficient in determining whether there were any objects in the working region.

The efficiency of the ultrasonic sensor also helped to make the system trigger more reliable because the system was able to differentiate between the presence and absence of objects.

Based on these observations, the ultrasonic sensor was selected as the final sensing component for system integration due to its material-independent detection capability and consistent performance.

3.1.3. Robotic System Performance

The performance of the robotic actuation mechanism was analyzed in relation to its capability to carry out sorting activities in real time following classification. Robotic manipulation with the use of MG996R servo motors was conducted continuously to measure movement precision and stability.

The robot effectively carried out the following tasks:

- Pick up of the object at its initial position
- Transporting it to the designated Region of Interest (ROI) for classification
- Showing the object to the user
- Final dropping into the proper classification category

Using an ROI eliminated the risk of inaccurate predictions due to noise or several objects in the picture.

All the joints were seen to be working very efficiently with very little mechanical errors or instabilities in their movements. There was not much of a delay between the results from the classification stage and actual action; hence, there was a high level of responsivity.

In general, the robot showed high levels of precision and stability, making it suitable for use in both automatic sorting and interactive learning purposes.

3.1.4. Integrated System Performance

The overall performance of the complete system was tested by integrating the vision system, sensor array, and robotics module in one process flow.

This system exhibited the following characteristics:

- Real-time accurate object identification with the DINOv2 model
- Successful detection and verification of objects via ultrasonic sensors
- Effective collaboration between perception and actuation
- Continuous operation without system failure

Overall, the entire process from receiving an object, processing it, and placing it in its correct position was done accurately, showing that integration of the entire system worked effectively. The whole process was also consistent throughout multiple trials, suggesting that the process is reliable.

In addition, adding an interactive step in which the robot shows the object to the student further improves the applicability of this project in an educational setting. Based on the findings, this project can be used as an intelligent robotic system capable of operating in real time.

3.2. Research Findings

This section presents the key insights derived from the experimental results, highlighting the broader contributions and implications of the research.

3.2.1. Effectiveness of Transformer-Based Models

One of the most prominent conclusions from this study is the remarkable ability of transformer-based self-supervised learning algorithms, including DINOv2, in recognizing textures of different materials.

In contrast to conventional CNNs, which mainly concentrate on local information, DINOv2 detects the global context of images and understands how to tell materials

apart based on their slight differences. This is crucial for fine-grained material texture recognition applications because even the slightest variation in textures may alter the outcome of the algorithm.

This work shows that transformer-based methods not only perform well in off-line image recognition tasks but can also be applied to real-time robotic perception systems.

3.2.2. Limitations of Classical CNN Architectures

Yet another noteworthy result was the inadequacy of using the traditional CNN-based models in tackling complex texture recognition scenarios.

It became evident that the Sequential CNN model could not perform well owing to its inefficiency in extracting features. Moreover, even though MobileNet and EfficientNet architecture proved to be better, they still demonstrated certain limitations when it came to working with similar visual traits of the materials being classified.

Hence, CNNs have shown themselves to be efficient tools in solving basic recognition problems, but they seem inadequate when it comes to material classification in dynamic conditions.

3.2.3. Importance of Multimodal Sensor Fusion

It was evident that the combination of visual classification with ultrasonic sensors was vital for enhancing the efficiency of the entire robot operation.

Although the vision classification model had been accurate, it was vulnerable to any environmental disturbances, including light intensity and position of the objects. In contrast, the inclusion of ultrasonic sensors provided a verification process by confirming that the robot would act only when there was an actual object.

This result confirms that multimodal perception systems have a significant role in robotics, where the combination of various sensory inputs results in enhanced accuracy.

3.2.4. Limitations of Capacitive Sensing

The assessment of the capacitive proximity sensor demonstrated major drawbacks in the application of such a sensor in a multi-material sensing system.

This inability to sense non-metallic materials proves its ineffectiveness in systems that involve multiple material types. This highlights the notion that material-specific sensors are not effective in universal robotic sensing systems.

3.2.5. Educational Impact of the System

Apart from its technical success, another important conclusion drawn from this research is the applicability of the system in an educational context.

The dynamic interaction process, whereby the robot introduces items to students, lets them predict the outcome, and gives instant feedback, results in a highly engaging learning process. This model facilitates active learning, where the learner is actively engaged in making decisions.

The system makes it easy for learners with various disabilities to learn and remember better. In addition, it helps demystify complicated AI theories by showing practical application.

This finding reaffirms the potential of combining AI and robotics in educating all individuals.

3.3. Discussion

The conclusions from this experiment confirm the success of integrating advanced artificial intelligence solutions, multi-modal sensing technologies, and robotic control into a single Woody collaboration platform that can perform material identification and classification in real time. In addition to its high efficiency, the system also proves the practicality of applying robotic perception and interactive learning algorithms, making it applicable in both automation and educational assistance purposes.

One of the major benefits of the developed Woody system includes the ability of using a self-supervised vision transformer: DINOv2 for identifying different materials based on visual textures. In contrast to CNNs, which apply hierarchical feature extraction focusing on local structures, this approach allows the use of attention to capture local and global relations. Consequently, the proposed algorithm performs well in detecting various types of materials, including those with similar textures. In other words, the system is capable of distinguishing visually similar objects such as plastic and rubber or glass and metal, as they may have common visual patterns in the context of different environments.

On the other hand, the paper brings forward the drawbacks of CNN-based architectures when used in texture recognition. The poor accuracy score (0.44) attained by the Sequential CNN implies that basic architectures are inefficient in recognizing textures. Although optimization strategies like EfficientNet and MobileNet provided better results with accuracies of 0.88 and 0.89 respectively, they were still inferior to the results achieved by DINOv2. It shows that while CNNs are computational lightweights ideal for small models, they might be inefficient when high-level tasks are required. Thus, future development of robot perception systems should focus more on transformer-based or hybrid approaches.

Sensing-wise, the choice to replace the capacitive proximity sensor with the ultrasonic sensor proved to be vital to enhancing the system's reliability. While the failure of the capacitive proximity sensor to sense objects made of nonmetallic materials indicated a weakness in material-based sensors, the ultrasonic sensor offered a solution by being able to reliably sense any objects irrespective of material composition. This

underscores the significance of understanding the need for choosing appropriate sensors based on the diversity of the environment during the development of robotic systems. The use of the ultrasonic sensor, in this case, supports the necessity of incorporating multimodal sensing into robot design.

The robotic actuation part of the system also proved to be quite efficient when tested for stability, response time, and precision. With the incorporation of MG996R servo motors, the robot could be used to execute the necessary steps for picking up the object, moving it into the specific ROI (region of interest), showing it to the user, and placing it down. The implementation of the ROI was important to ensure that the recognition process would not be disturbed by noise or other objects, hence increasing the accuracy of predictions. Another important factor was that the system had low latency, which was necessary for any real-time application scenario.

One of the key strengths of the research conducted is related to the development of a model for interactive human-robot collaboration, wherein the robotic device shows the object to the user prior to carrying out the final sorting step. In such a way, it is possible to turn the system not only into the effective device for sorting objects but also into the tool capable of fostering a higher degree of participation in learning process, as well as promoting involvement and interaction in the classroom environment. This aspect correlates with the ideas about the necessity of implementing practices related to participatory learning.

The application of this kind of the robotic system can prove helpful in the field of assistive education, as many differently-abled students face issues when learning in the classroom due to abstractness of the traditional learning strategies used by instructors. With the help of the proposed system, it is possible to integrate various components into the learning process, including cognitive aspects (decision making), visual component (classification made by the algorithm), and physical interaction (interactions with the robotic device).

However, there are some constraints within the study that need to be recognized. First, the dataset employed for training is extensive but confined to a set number of material types. Adding more material samples and testing under different environmental

scenarios will further augment the performance of the algorithm. Second, even though DINOv2 provided highly accurate predictions, its heavy computation process makes it difficult to implement in resource-constrained computing devices compared to other models. Future research can look into optimizing the model to make it efficient without losing too much accuracy.

Furthermore, the current system is configured to use a single robotic arm with limited operations. Developing a system that detects multiple objects and operates in dynamic environments will expand its applications both in industrial settings and in learning institutions.

In conclusion, the analysis shows that the combination of self-supervised vision transformers, multimodal sensors, and the mechanisms for cooperation between humans and robots provides a solid foundation. Not only does the solution show high-level efficiency in material identification and sorting using robotics, but it also proves applicable within the domain of assistive education. In other words, there is a dual value to the suggested method which emphasizes its importance.

4. CONCLUSION AND FUTURE WORK

4.1. Conclusion

The objective of this research was to design and develop an adaptive human–robot collaborative system, Woody, capable of performing intelligent material texture classification while also serving as an assistive educational tool for differently-abled students. The system integrates deep learning-based vision models, multimodal sensor fusion, and robotic actuation into a unified framework that enables real-time perception, decision-making, and physical interaction.

The results of this study clearly demonstrate that the proposed system successfully achieves its intended goals. The vision-based classification module, powered by DINOv2, achieved a high accuracy of approximately 98%, significantly outperforming traditional convolutional neural network architectures such as Sequential CNN, MobileNet, and EfficientNet. This confirms that self-supervised vision transformers are highly effective in capturing fine-grained texture differences between materials such as plastic, glass, rubber, and metal.

The sensor subsystem further improved system reliability by introducing an additional layer of environmental awareness. Although the capacitive sensor was initially explored, its limitations in detecting non-metallic materials reduced its effectiveness in a multimaterial context. The integration of the ultrasonic sensor resolved this issue by providing consistent object detection and spatial awareness, ensuring stable operation of the robotic system.

The robotic actuation system demonstrated accurate and real-time execution of sorting tasks using servo-based control mechanisms. The system maintained synchronization between perception and physical action, enabling smooth object handling and placement into designated categories.

Beyond technical performance, a significant outcome of this research is its application in assistive education. The system transforms complex artificial intelligence processes into observable physical actions, allowing learners—particularly differently-abled students—to engage with and understand material classification concepts in an interactive and meaningful way. Through collaboration with SERD Kelaniya, the

system demonstrates strong potential for deployment in inclusive educational environments.

In conclusion, this research successfully bridges the gap between artificial intelligence, robotics, and inclusive learning, providing both a technically robust and socially impactful solution.

4.2. Contributions of the Research

This research makes several important contributions to the fields of human–robot collaboration, computer vision, and assistive educational technology:

Firstly, it demonstrates the effectiveness of self-supervised vision transformers (DINOv2) in real-time robotic material classification tasks, proving their superiority over conventional CNN-based architectures.

Secondly, it presents a multimodal fusion approach combining vision and ultrasonic sensing to improve system robustness and reduce uncertainty in real-world robotic operations.

Thirdly, it introduces a human-centered robotic system design, where robotics is not only used for automation but also as a medium for education and interaction.

Finally, it contributes to the development of assistive learning technologies for differently-abled students, showing how advanced AI systems can be adapted for inclusive education.

4.3. Limitations of the Study

Despite the successful implementation of the system, several limitations were identified during the research.

One major limitation is that the system is currently restricted to four material categories. Expanding the dataset to include a wider range of materials such as wood, fabric, and ceramics would improve generalization and practical usability.

Another limitation is the dependency on controlled environmental conditions during testing. Although the system performs well under varying lighting conditions, extreme environmental variations such as strong reflections or cluttered backgrounds may still affect classification accuracy.

Additionally, the robotic system is limited in terms of mechanical complexity. The current servo-based arm is suitable for demonstration purposes but may not be sufficient for high-speed industrial applications.

4.4. Future Work

Future improvements of this research can be explored in several directions to enhance both technical capability and real-world applicability.

4.4.1. Expansion of Material Categories

Future versions of the system can include a broader range of materials beyond the current four classes. This would make the system more applicable to real-world sorting environments such as recycling facilities and industrial automation systems.

4.4.2. Integration of Advanced Sensor Technologies

The current system uses ultrasonic sensing for object detection. Future work can explore the integration of more advanced sensors such as:

- Infrared depth sensors
- 3D LiDAR systems
- Hyperspectral imaging sensors

These additions would significantly improve environmental perception and material differentiation capability.

4.4.3. Improvement of Robotic Intelligence

The robotic control system can be enhanced by incorporating reinforcement learning-based control strategies. This would allow the robotic arm to learn optimal movement strategies over time, improving efficiency and adaptability in dynamic environments.

4.4.4. Real-Time Edge Deployment

Although the current system performs well in a controlled environment, future work can focus on deploying the model on edge computing devices. This would enable fully autonomous operation without reliance on external computing resources, improving portability and scalability.

4.4.5. Enhanced Assistive Learning Features

Future development can focus on making the system more interactive for educational purposes. Possible enhancements include:

- Voice-based explanations of classification results
- Visual feedback systems for learners
- Adaptive difficulty levels based on student interaction

These improvements would further strengthen the system's role in inclusive education.

4.4.6. Multi-Robot Collaborative Systems

A long-term extension of this work could involve multiple robotic agents working collaboratively to perform sorting tasks. This would better simulate real industrial environments and introduce more complex human–robot collaboration scenarios.

4.5. Final Remarks

This research demonstrates that combining advanced AI models, sensor fusion, and robotics can produce intelligent systems that are not only technically effective but also socially meaningful. The Woody system serves as a foundation for future developments in both intelligent automation and assistive education, highlighting the importance of designing technology that benefits a broader range of users, including differently-abled individuals.

The integration of DINOv2-based perception with real-time robotic execution represents a significant step toward next-generation human–robot collaborative systems that are adaptive, accurate, and inclusive.

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